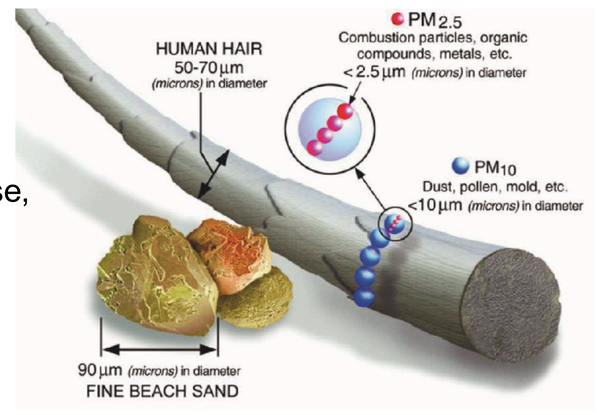
Design and Implementation of an IoT-Enabled Air Quality Monitoring and Recommendation System with Mobile App Integration for Urban Areas

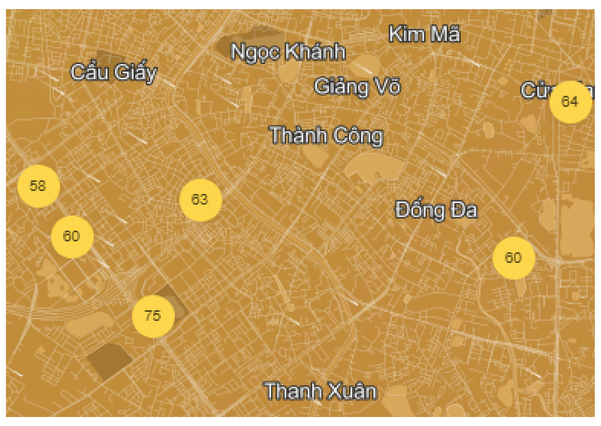
Duc Minh Vu[1], Anh Duc Vu[2], Quang Minh Bui[2]

[1]Hanoi-Amsterdam High School for The Gifted
[2]HUS High School for Gifted Student

Introduction

- PM2.5 (< 2.5μm) penetrates deep into the lungs and bloodstream.
- WHO: **~7 million deaths**/year linked to air pollution.
- PM2.5 = Group 1 carcinogen (heart disease, stroke, lung cancer).





Hanoi: High Risk, Low Awareness

- PM2.5 levels exceed WHO limits by 4–5× during dry seasons.
- Many low-income communities are unaware of the invisible threat.
- Public air quality data is often incomplete or hard to access.

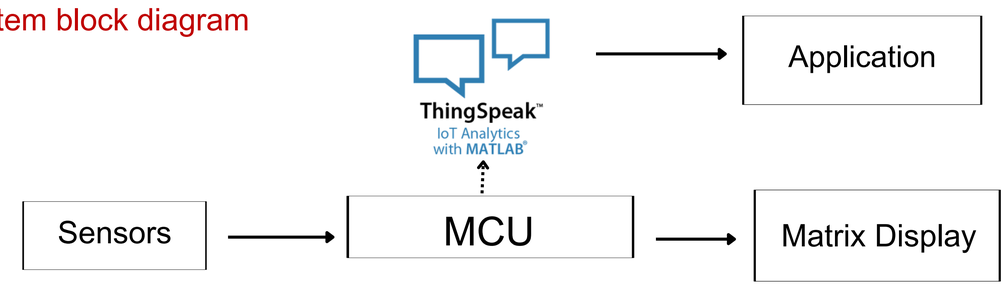
Our project objectives are a low-cost, solar-powered air quality system with real-time display, mobile app, and AI forecasting.

System overview

Our system provides

- Real-time data on public LED boards at urban sites (e.g., schools, hospitals, markets)
- Automated alerts issued to vulnerable populations during pollution spikes
- High-pollution prediction using AI models for accurate urban forecasting

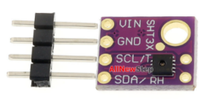
System block diagram



Hardware



SDS011



SHT30

LILYGO T-A7670 ESP32



WS2812B2 LEDs



System Intergration



Sensors

SDS011 PM2.5/PM10 Sensor

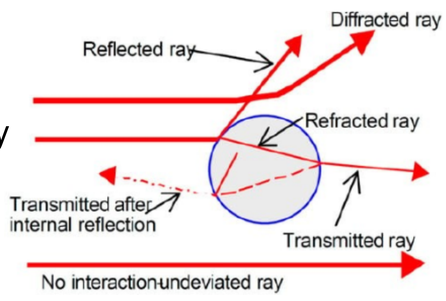
- **Low-cost** laser sensor for real-time PM2.5/PM10 monitoring
- Uses laser scattering via photodiode
- Ideal for low-cost environmental/health research

SHT30 Humidity, Temperature Sensor

- Digital sensor for **accurate** temperature and humidity monitoring
- I²C Communication; **easy to integrate**
- Low power, fast response, and stable over time

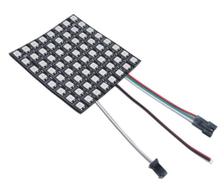
MCU — LILYGO TTGO T-A7670G ESP32

- ESP32-based board with LTE (A7607G)
- Controls LED matrix for real-time air quality display
- Sends data to ThingSpeak via HTTP over LTE
- Supports **low-power, wireless, cloud-connected** monitoring



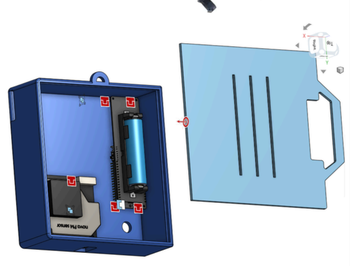
LED Display — 4x WS2812B2 LED

- RGB LED matrix with built-in drivers, controlled via 1 GPIO
- 4 panels daisy-chained, driven by ESP32 (LILYGO)
- Displays **real-time air quality data** as RGB color codes



3D-printed weatherproof box from recycled PLA

- Custom-designed in Onshape and 3D-printed from recycled PLA for a **weatherproof, eco-friendly** enclosure



Solar Panel

- Provides **off-grid**, renewable power for continuous system operation
- Enables sustainable, autonomous environmental monitoring without reliance on external power sources

System Intergration

The hardware system integrates sensors and display components onto a single ESP32 platform, ensuring synchronized operation via UART and I²C protocols. Addressing and communication channels were carefully assigned to prevent data collisions.

Software

MCU — Software Structure

boot.py → Initializes system
main.py → Starts main loop or OTA update
loop_sds011.py → Runs sensor logic and data handling
sds011.py → Controls SDS011 dust sensor via UART; handles data parsing, sleep/wake commands
ledpm25.py → Manages 64x16 LED matrix display; scrolls PM2.5, PM10, temperature, humidity
sim7600.py → Interface with GSM module; send HTTP request to ThingSpeak

Main Loop Workflow

Wake SDS011, wait 5 second
↓
Read PM2.5 / PM10
↓
Read Temp&Humidity from SHT30
↓
Display data on LED matrix
↓
Send to ThingSpeak via GSM
↓
Put Sensors to sleep
↓
Repeat

Thingspeak Cloud Server

1.Sensor Data Collection
→ PM2.5, PM10 (SDS011)
Temperature, Humidity (SHT30)

→

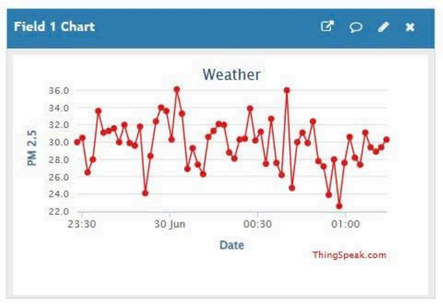
2. URL Construction (HTTP GET)
→ Data is formatted into a URL with unique API

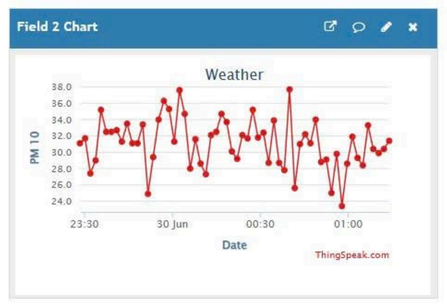
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3. Data Transmission via GSM
→ SIM7600 module connects to the cellular network, sends data by HTTP GET request

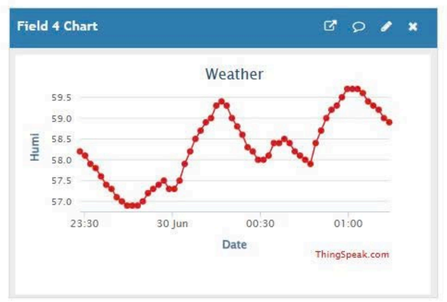
←

4. Server Response
→ ThingSpeak returns an entry ID on success and vice versa







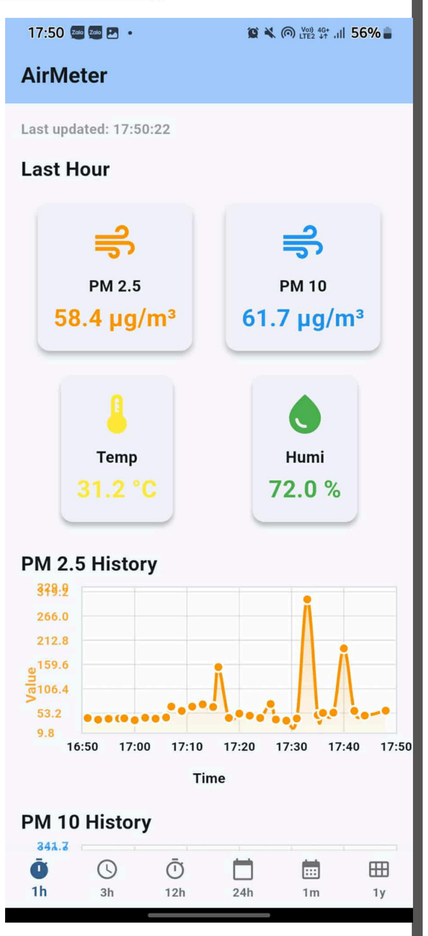


Thingspeak Public Channel

App Download File

Mobile Application

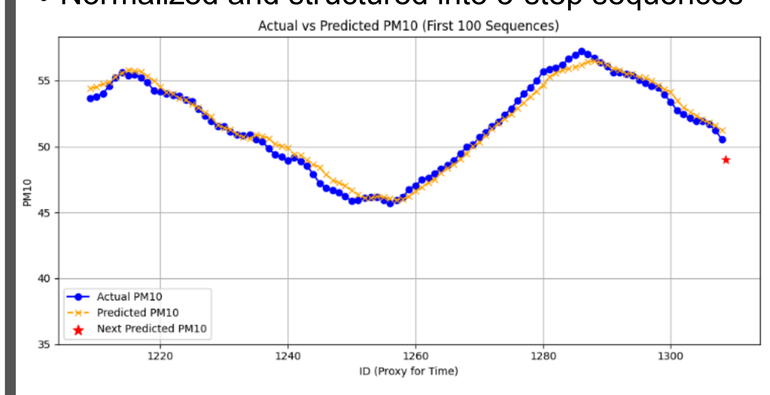
- **Real-time /historical data** from ThingSpeak via HTTP REST API
- **Interactive charts** using fl_chart for clear time-series visualization
- **Custom time ranges** (1 hour to 1 year) with auto-aggregation
- **Threshold alerts f**or immediate notification of critical pollutant levels
- **Auto-refresh** every 10 seconds for near real-time updates
- **Remote access** promotes community awareness and data-driven action

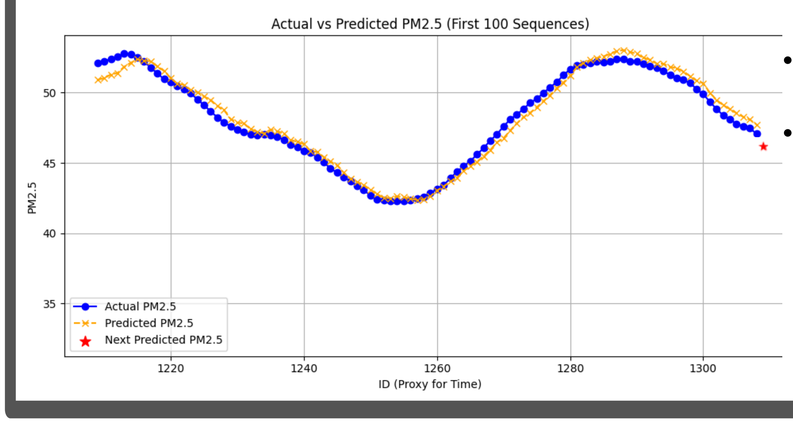


AI Forecasting Model

1. Input & Preprocessing

- Real-time sensor data: PM2.5, PM10, temperature, humidity
- 30-minute rolling averages (15 readings)
- Normalized and structured into 3-step sequences





2. Model Architecture

- 1D Convolutional Neural Network (CNN)
- Layers: Conv1D → MaxPooling → Flatten → Dense
- Output: 4 predicted values for next 30-minute window

3. Results & Prediction

- Accurately forecasts short-term air quality
- Smooth training and validation loss over 100 epochs

AI Model link (google collab)

ADVANTAGES

- Real-time LED display
- Compact, low-power, solar
- Weatherproof 3D-printed case
- AI-based forecasts
- Mobile app with live data

CONCLUSION

- A low-cost, eco-friendly solution for air quality monitoring tailored to urban schools and neighborhoods.
- Integration of IoT and AI enables timely data collection, forecasting, and awareness building.

FUTURE PLAN

- Improve AI accuracy by integrating real-time weather and traffic data.
- Development of alerts and recommendation
- Add more sensors (CO, NO_2...)
- Explore integration with local health and education authorities

SUSTAINABILITY COMMITMENT

- Source code on [GitHub] and box CAD on [Onshape]—freely available for reuse, adaptation, and expansion.

Box CAD File ***Github Repository***